

## 1 Question (10 points)

Let  $A_n = \{a^k \mid k \text{ is a multiple of } n\}$ . Show that for each  $n \geq 1$ , the language  $A_n$  is regular.

## 2 Question (10 points)

Let  $B_n = \{x \mid x \text{ is a binary number that is a multiple of } n\}$ . Show that for each  $n \geq 1$ , the language  $B_n$  is regular.

### 3 Question (10 points)

Is the following language context-free? Prove your answer, either by pumping lemma or by giving CFG. You are not allowed to use PDA.

$$C = \{a^n b^m c^m d^n e^p f^p \mid n, m, p \geq 0\}$$

## 4 Question (10 points)

Give PDA that generates the following language. The alphabet is  $\{0, 1\}$ .

$D = \{w \mid w \text{ contains at least two 0s and one 1}\}$

## 5 Question (10 points)

Design a single tape TM that recognizes the language  $L = \{0^n 1^m 2^p 3^{n+m+p} | n, m, p \geq 1\}$ . Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.